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PSAC Exam Practice — Mauritius

2026

Grade 6 • Science

Specimen Paper 3

Time allowed: **1 hour 45 minutes**

Total marks: **100**

Candidate Name: _____

School: _____

Date: _____

Index No.: _____

Instructions to Candidates

1. Write your name, school and index number clearly on this paper.
2. Answer **ALL** questions in **ALL FOUR** sections.
3. Write your answers in the spaces provided on this paper.
4. All diagrams must be neat, clearly labelled and drawn in pencil.
5. The marks for each question are shown in square brackets [].
6. Calculators and electronic devices are **NOT** permitted.
7. Where working is required, show all steps clearly.

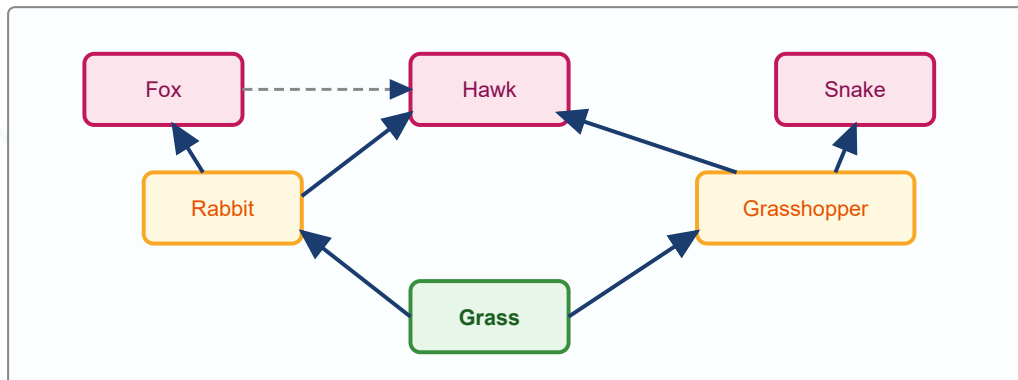
Section	Topic	Available	Awarded
A	Living Things — Ecosystems & Adaptation	30	
B	Matter — Properties of Materials	25	
C	Energy — Forces & Simple Machines	25	
D	Earth & Space — Rocks & Soil	20	

Section	Topic	Available	Awarded
TOTAL		100	

SECTION A — Living Things: Ecosystems & Adaptation [30 marks]

1 The diagram below shows a food web in a grassland ecosystem.

[10]



(a) Using the food web, write **two** complete food chains that end with the Hawk.

[2]

1.

2.

(b) What is a food web? How does it differ from a food chain?

[2]

(c) If a disease killed all the snakes, predict what would happen to the **grasshopper** population and to the **hawk** population. Explain your reasoning.

[4]

- (d) Name the type of organism that breaks down dead plants and animals and returns nutrients to the soil.

[1]

- (e) The arrows in a food web represent the flow of something. What do the arrows show?

[1]

- 2 This question is about habitats and how organisms are adapted to survive.

[10]

- (a) What is a *habitat*? Give one example of a habitat found in Mauritius.

[2]

- (b) For each organism, describe **one** structural adaptation and explain how it helps the organism survive.

[6]

Organism	Adaptation	How it helps
Camel		
Polar bear		
Cactus		

- (c) What is the difference between a *structural* adaptation and a *behavioural* adaptation? Give one example of each.

[2]

3 This question is about environmental impact on ecosystems.

[10]

(a) State **three** human activities that can damage ecosystems.

[3]

1.

2.

3.

(b) The dodo was a bird found only in Mauritius. It became extinct in the 17th century. Suggest **two** reasons why the dodo became extinct.

[2]

(c) What does *biodiversity* mean, and why is it important for ecosystems?

[2]

(d) Describe **two** ways in which we can protect endangered species in Mauritius.

[2]

(e) What is a *carnivore*? Give one example from the food web in Question 1.

[1]

SECTION B — Matter: Properties of Materials [25 marks]

4 This question is about the properties of everyday materials.

[9]

- (a) Complete the table by filling in a suitable material and explaining why it is chosen for each use.

[6]

Object	Material Used	Property that Makes it Suitable
Window pane		
Electrical wire insulation		
A bridge		

- (b) State the difference between a *conductor* of heat and an *insulator* of heat. Give one example of each.

[3]

--

5 This question is about changes to materials.

[8]

- (a) Classify each change as *reversible* or *irreversible*. Give a reason in each case.

[6]

Change	Rev. or Irrev.?	Reason
Ice melting to water		
Wood burning to ash		
Cake baking in an oven		

- (b) Iron rusts when left in damp air. Is rusting reversible or irreversible? Name the two substances (other than iron) needed for rusting to occur.

[2]

--

6 This question is about natural materials and their uses.

[8]

- (a) Name **three** natural materials we obtain from the Earth and state one use for each.

[3]

Natural Material	One Use

- (b) What is the difference between a synthetic material and a natural material? Give one example of each.

[2]

- (c) Plastic does not rot (it is non-biodegradable). Why is this a problem for the environment? Suggest **one** way to reduce plastic waste.

[3]

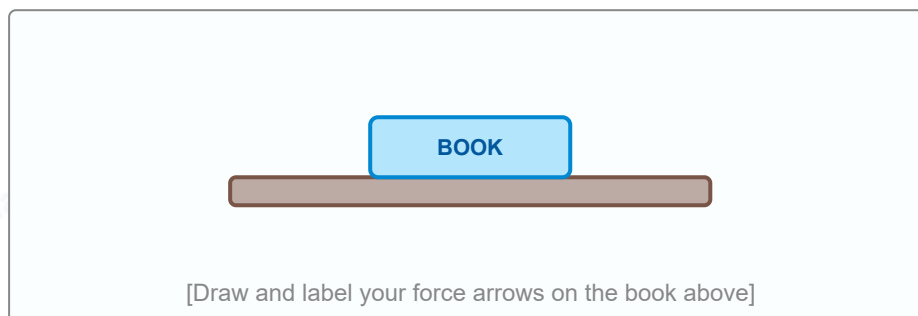
SECTION C — Energy: Forces & Simple Machines [25 marks]

7 This question is about forces.

[9]

- (a) The diagram below shows a book resting on a table. Draw and label **two** forces acting on the book in the space provided.

[2]



- (b) Match each description to the correct type of force.

[4]

Pulls objects towards the Earth

Gravity

Opposes motion between two surfaces

Friction

Pulls magnetic materials

Magnetic attraction

Pushes objects apart (like poles)

Magnetic repulsion

- (c) A pupil pushes a book across a rough table and it slows down. Explain what causes the book to slow down and state whether friction is useful or harmful in this situation.

[2]

- (d) The unit of force is the Newton (N). If an object weighs 50 N on Earth, what causes this weight?

[1]

8 This question is about simple machines that make work easier.

[9]

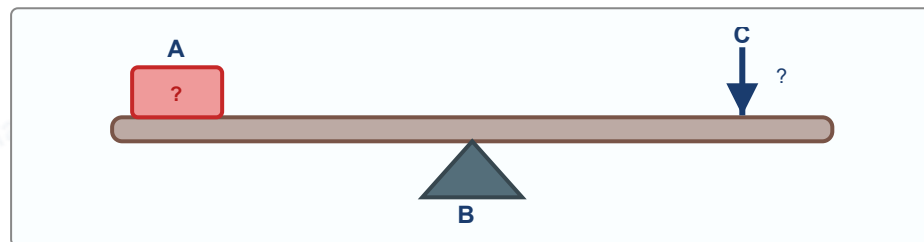
(a) Name the type of simple machine shown in each example below.

[4]

Example	Type of Simple Machine
A see-saw (plank balanced on a pivot)	
A ramp used to load a heavy box into a truck	
A flagpole with a rope and wheel at the top	
A wheel on a bicycle hub (axle)	

(b) The diagram shows a lever. Label the fulcrum, the effort, and the load.

[3]



A: _____ B: _____ C: _____

(c) A long lever arm helps you lift a heavy rock with less effort. Explain why, using the idea of moments (turning force).

[2]

9 This question is about pressure and upthrust.

[7]

(a) What is pressure? Write the formula for calculating pressure.

[2]

(b) A block has a weight of 60 N. It has a base area of 0.03 m². Calculate the pressure it exerts on the floor. Include the unit in your answer.

[3]

- (c) A steel ship floats on water even though steel is denser than water. Use the idea of *upthrust* to explain why.

[2]

SECTION D — Earth & Space: Rocks & Soil [20 marks]

10 This question is about the three types of rocks.

[10]

(a) Complete the table about the three types of rock.

[6]

Rock Type	How it Forms	Example
Igneous		
Sedimentary		
Metamorphic		

(b) What is the rock cycle? Briefly explain how igneous rock can become sedimentary rock over time.

[3]

(c) What is meant by *weathering* of rocks? Name **two** agents of weathering.

[1]

11 This question is about soil and its importance.

[10]

(a) Name **four** components (parts) of soil.

[4]

1. _____

2.

3. _____

4.

- (b) A pupil tests three soil samples for drainage using the same amount of water. The results are shown below.

Soil Sample	Time for water to drain through
Sandy soil	1 minute
Clay soil	25 minutes
Loam (garden soil)	6 minutes

- (i) Which soil drains most slowly? Explain why.[2]

- (ii) Which soil would be best for growing most crops? Give a reason.
[2]

- (c) What is *soil erosion*? Suggest **one** way to prevent soil erosion on farmland.
[2]

Mark Scheme — Specimen Paper 3 (2026) • Grade 6 Science

For teachers and parents: Accept any answer that demonstrates equivalent scientific understanding. Where wording differs, award the mark if the key concept is clearly expressed. This paper is for practice purposes only. Produced by Noukclass.com.

Q	Part	Acceptable Answer(s)	Marks
1	(a)	Any two valid chains: e.g. Grass → Rabbit → Hawk; Grass → Grasshopper → Hawk. [1 each]	2
	(b)	A food web shows all the feeding relationships between organisms in an ecosystem. A food chain shows only one pathway; a web shows multiple interconnected chains. [1+1]	2
	(c)	Grasshopper: would increase (no snakes to eat them) [1+reason 1]. Hawk: likely to increase initially (more grasshoppers to eat) [1+reason 1].	4
	(d)	Decomposers (bacteria/fungi/microorganisms)	1
	(e)	The flow of energy (from one organism to the next)	1
2	(a)	Habitat = the natural environment where an organism lives and finds food, shelter, water. Example: coral reef / mangrove forest / sugarcane field / forest. [1+1]	2
	(b)	Camel: hump (stores fat for energy/water), large feet (spread weight on sand), long lashes (protect eyes from sand). Polar bear: thick white fur (insulation + camouflage), layer of blubber (warmth). Cactus: thick waxy stem (stores water), spines instead of leaves (reduce water loss). [1 adaptation + 1 reason per organism, max 6]	6
	(c)	Structural = a physical feature of the body (e.g. thick fur). Behavioural = something the animal does (e.g. hibernation, migration). [1 each + 1 example each, max 2]	2
3	(a)	Any three: deforestation, pollution, overfishing, hunting/poaching, urbanisation, introduction of invasive species, climate change. [1 each]	3
	(b)	Any two: hunted by humans (and introduced animals like pigs/rats); habitat destruction; could not fly to escape predators; laid only one egg at a time. [1 each]	2
	(c)	Biodiversity = the variety of living species in an ecosystem [1]. Important because: greater variety means the ecosystem is more stable and resilient; if one species is lost, others can fill its role [1].	2

Q	Part	Acceptable Answer(s)	Marks
	(d)	Any two: create nature reserves/national parks; breeding programmes in captivity; ban hunting/poaching; protect habitats from development. [1 each]	2
	(e)	Carnivore = an animal that eats only other animals. Example from web: Fox, Hawk, Snake. [1]	1
4	(a)	Window pane: glass; transparent (allows light through). Electrical wire insulation: plastic/rubber; electrical insulator (does not conduct electricity). Bridge: steel/concrete; strong, rigid, high tensile strength. [2 per row correct: material + property]	6
	(b)	Conductor: allows heat to flow through easily (e.g. metal/copper). Insulator: does not allow heat to flow easily (e.g. wood, plastic, wool). [1 per correct definition + 1 per example, max 3]	3
5	(a)	Ice melting: reversible (water can be frozen again). Wood burning: irreversible (ash cannot become wood again). Cake baking: irreversible (cannot un-bake a cake). [2 per row]	6
	(b)	Irreversible. Substances needed: water and oxygen (air). [1+2]	2
6	(a)	Any three: rock/stone (building), iron/metal (tools), clay (pottery/bricks), sand (glass), coal (fuel), oil (plastic/fuel), limestone (cement). [1 per correct pair]	3
	(b)	Natural material: found in nature, not made by humans (e.g. wood, cotton). Synthetic: made by humans/in a factory (e.g. plastic, nylon). [1 per correct definition + example]	2
	(c)	Problem: plastic stays in the environment for hundreds of years; harms animals that eat it or get tangled; pollutes oceans and land [1]. Reduction: use reusable bags/bottles; recycle plastic; ban single-use plastics [1]. [1 per correct idea, max 2 + 1 for suggestion]	3
7	(a)	Two arrows on book: downward arrow labelled "Weight/Gravity" and upward arrow labelled "Normal reaction/Support force". [1 each]	2
	(b)	Pulls towards Earth → Gravity. Opposes motion → Friction. Pulls magnetic materials → Magnetic attraction. Pushes apart → Magnetic repulsion. [1 each]	4
	(c)	Friction between the book and table opposes the motion, slowing the book down [1]. In this case friction is harmful (it stops the book moving) [1].	2
	(d)	Gravity (the gravitational pull of the Earth)	1
8	(a)	See-saw: lever. Ramp: inclined plane. Flagpole: pulley. Wheel on hub: wheel and axle. [1 each]	4
	(b)	A = Load, B = Fulcrum, C = Effort. [1 each]	3

Q	Part	Acceptable Answer(s)	Marks
	(c)	Moment = force × distance from fulcrum. A longer effort arm increases the moment produced by a smaller effort force, so less effort is needed to balance (lift) the same load. [1+1]	2
9	(a)	Pressure = the force acting per unit area. Formula: Pressure = Force ÷ Area ($P = F/A$). [1+1]	2
	(b)	$P = 60 \text{ N} \div 0.03 \text{ m}^2 = \mathbf{2000 \text{ Pa (N/m}^2\text{)}}$. [1 for correct substitution, 1 for correct calculation, 1 for unit]	3
	(c)	The hollow shape of the ship displaces a large volume of water; the upthrust (upward force from the water) equals the weight of water displaced. If this upthrust equals the ship's weight, the ship floats. [1+1]	2
10	(a)	Igneous: forms when molten rock (magma/lava) cools and solidifies; example: granite, basalt. Sedimentary: forms from layers of sediment compacted over time; example: sandstone, limestone. Metamorphic: forms when existing rock is changed by heat and pressure; example: marble, slate. [2 per row]	6
	(b)	Rock cycle = the continuous process by which rocks are formed, broken down and reformed [1]. Igneous rock is weathered by wind/rain into small particles → transported and deposited in layers → compressed over time to form sedimentary rock [1 per step, max 2].	3
	(c)	Weathering = the breakdown of rocks into smaller pieces. Agents: water/rain/frost action, wind, temperature changes, acid rain, tree roots. [1 for definition, 2 agents = 1 mark]	1
11	(a)	Any four: rock particles/minerals, humus (decayed organic matter), water, air/gases, living organisms. [1 each]	4
	(b)(i)	Clay; clay particles are very small and packed tightly together, so there are tiny air spaces and water cannot drain quickly. [1+1]	2
	(b)(ii)	Loam; it drains well enough not to waterlog but retains enough moisture and nutrients for plant growth. [1+1]	2
	(c)	Soil erosion = when topsoil is worn away by wind or water [1]. Prevention: plant trees/grass to hold soil; terracing on slopes; use windbreaks [1].	2
TOTAL			100